

Is a Fully Self-Sufficient Economy Still Possible?

Zhihu · 2026-09-18

Since Trump launched his trade war, articles on de-globalisation or the disintegration of globalisation have been everywhere, giving the impression that the global economy is bound to move towards a state in which each region, and even each country, runs on its own and in isolation from the others. But the reality is that although FDI has indeed fallen for political reasons, it remains above \$1 trillion^{[1][2]}

There are many reasons why the trend of globalisation is hard to reverse completely, and no single article can set them all out. In this article we shall analyse the goal state of de-globalisation: is it possible for each country to run a completely self-sufficient economy on its own?

The Population Constraint

People are the most central element of the economy, so we begin by discussing the lower limit of manpower for a self-sufficient economy — how many people are needed at a minimum to run an independent economy. On this question I have found no paper that analyses it head-on, perhaps because the modern economy is too complex: to establish the minimum population requirement one needs a great deal of real economic data, and often much of it is missing. A single industry in national statistics may cover many different products, and producing those products may in turn require factories of varying numbers. To analyse, in an already globalised world, the minimum size of an economy that sustains the same level of consumption and the same variety of products is therefore extremely difficult, even if it were possible. The only approach, then, is to look at countries that have historically been in a self-sufficient state.

Let us leave aside the nineteenth century. In the twentieth century, the economies that were at one time self-sufficient were the following: 1) Nazi Germany in the Second World War, 2) Japan in the Second World War, 3) the United States in the Second World War, 4) the Soviet and Eastern European socialist camp during the Cold War, 5) the capitalist industrial countries of the Cold War — the system of capitalist industrial countries formed by the US, Europe, Japan, Korea and Canada. Of these, 1), 2) and 3) were industrial societies at the level reached after the Second Industrial Revolution, while 4) and 5) were societies after, or in the course of, the Third Industrial Revolution. Let us look at the populations of a few of these economies.

The total population of Nazi Germany at the 1939 census was nearly 80 million^[3]

During the Second World War, although the American labour market was tight, there was no shortage as severe as that in Germany or Japan. American industrial goods had to supply not only its own army but also allies such as the Soviet Union, Britain and China. And since the other industrial countries were at war, and neutral countries such as those of Latin America had very low industrial levels, they could not supply much to the United States. We may therefore take it that the United States in the Second World War was able to sustain a self-sufficient economy.

In 1945 the total population of the United States was nearly 140 million^{[4][5]} — between 80 million (Nazi Germany) and 140 million (the United States). Indeed, back in the 1970s some scholars held that 20 million people were not enough to establish a self-sufficient economy, but that 100 million were not needed either^{[6][7]}

Of course, because the Third Industrial Revolution multiplied the kinds of industries in human society and made the division of labour finer, the population requirement has risen higher.

In 1985 the total population of the whole Soviet and Eastern European region was about 380 million^{[8][9]}

During the Cold War the West imported mainly oil and gas from the Soviet bloc, while the Soviet and Eastern European countries needed to import advanced equipment from Europe, America and Japan. We therefore cannot be certain whether the total population of the Soviet bloc was enough to complete the whole range of industries, but we can be certain that the industries of the Western camp were essentially complete. For a self-sufficient economy in the course of the Third Industrial Revolution, then, the population requirement lies between 400 million (the Soviet bloc) and 700 million (the capitalist industrial countries).

Combining these two figures, let us see which countries in the world can meet the two requirements.

Countries with more than 80 million people: India, China, the US, Indonesia, Pakistan, Nigeria, Brazil, Bangladesh, Russia, Mexico, Japan, the Philippines, the Democratic Republic of the Congo, Ethiopia, Egypt, Vietnam, Iran, Turkey and Germany. Among them, India, China, the US, Indonesia, Pakistan, Nigeria, Brazil, Bangladesh and Russia have more than 140 million people, exceeding what the Second Industrial Revolution required and leaving them energy to develop other industries. So fewer than 10 countries in the world are qualified even to "consider" self-sufficiency. Their total population is about half of humanity. That is to say, even if they all went it alone, nearly half of mankind could not do so and would have to depend on trade with one another.

Countries with more than 400 million people number only two — China and India; and if the EU counts as one economy, then the EU (450 million) also meets the standard. Nigeria may in future also reach 400 million. As for the 700 million that is the safer lower limit for a Third-Industrial-Revolution economy, only China and India have ever had a population of that size in human history.

As population declines and ages, China's working-age population will fall below 400 million by the 2070s, and its total employed population will be lower than the total employment of the Western industrial countries in 1985, so it will not meet this standard. Yet even if both China and India chose to go it alone, they would take with them only a third of the world's population; the other two thirds would still have to trade and cooperate in the division of labour with one another.

In fact, after 1985 the division of labour in production continued to develop further, raising the population requirement still higher. China, as an example of something close to a full industrial chain, can be used to analyse the population requirement of a self-sufficient economy today. In 2025 China's non-agricultural working population was nearly 600 million. China's manufacturing exports were 25% to 30% of manufacturing GDP, and total exports about 20% of GDP. Even taking the high figure of 30%, the share of Chinese workers' total labour that serves China itself reaches 70%. China therefore has 420 million non-agricultural workers serving the domestic economy; adding, on the assumption of agricultural modernisation, 30 million more agricultural workers (roughly the same as China's total agricultural employment in 2050^[10]) the lower limit for the total labour force is 450 million, which, on an 80% labour participation rate and a 60% working-age share of the population, implies a total population of about 940 million, of whom 560 million would be of working age. On current population trends, China's working-age population will fall below this level in the 2060s. On the UN's pessimistic projection, that shift will be complete in the 2050s. Whether the standard is 700 million or 940 million, in the second half of the twenty-first century only India will meet the requirement.

Natural Resources

Besides population, natural resources are a key constraint. Among the countries and regions that reach 400 million people, the EU lacks oil and gas, as well as critical minerals such as rare earths, lithium, bauxite, copper and cobalt. India likewise lacks oil and gas, and the copper, lithium and other minerals that advanced industries need; see "India's Natural Resources and Its Industrialization" for details. China's position is much better than either: although it must import large quantities of iron ore, copper, oil and gas, its domestic reserves are in fact considerable. The crux is minerals such as chromium, manganese, cobalt, lithium and nickel, where domestic reserves are low and imports are relied on heavily.

With the rise of resource nationalism, resource countries are one after another banning the export of raw ore and demanding local processing. An independent industrial country is therefore obliged to develop a certain export industry in order to earn foreign exchange and import processed mineral products. And because local processing raises value added, the cost of the imports it needs will go on rising compared with importing raw ore. Take lithium: China imports more than \$5 billion of lithium

a year, and its copper imports run above \$150 billion, while imports of various other metals are not small either. India now spends more than \$10 billion a year importing copper. As a country industrialises and resource countries process more at home, import costs will keep increasing.

If a few large countries move towards self-sufficiency and industrial chains are cut off, then as standards and interfaces evolve, it is quite likely that their products will cease to fit with those of other countries, leaving them able to export only mass consumer goods and primary resource products. And once industrial chains are decoupled, any economic fluctuation can easily bring about a drying-up of foreign exchange.

Small and medium-sized countries cannot take the road of self-sufficiency because of their limited populations, so they will continue to uphold the trading system and, through their possession of natural resources, force the large countries to take part in the global division of labour in trade.

The Technological Constraint

Because each of the three industrial revolutions has been more complex than the last, it has become harder and harder for late-developing countries to catch up. In the First Industrial Revolution, an engineer or skilled worker who slipped away abroad could reproduce the machines in a British factory. Catching up was therefore easy for late developers and took little time. Germany, for example, began its industrial revolution in the 1830s and had basically completed it by the 1860s — about 30 years from start to finish.

After the Second Industrial Revolution, however, not only was the technology more complex, but the capital and the scale of skilled workers required were greater, so catching up took far longer. Japan, for example, spent about 60 years building a complete modern industrial system between the Meiji Restoration and the eve of the Second World War.

After the Third Industrial Revolution, no complete latecomer has yet caught up — not one. Only China in the whole world is in the course of catching up: it will be able to complete the building of an independent industrial system in the 2030s and reach the world's advanced level in the 2040s. The reason is that the Third Industrial Revolution unfolded within a global division of labour, and it is extremely difficult for any single country to assemble an independent industrial system. Only a country with China's population size can even contemplate so vast an industrial plan.

For the future economic giants — India, Pakistan, Bangladesh, Nigeria, Ethiopia, Egypt and the Democratic Republic of the Congo — therefore, developing advanced industries such as semiconductors and large aircraft can only begin with entering the global division of labour and finding a few sub-sectors in which to concentrate. Of course, 30 years from now India may well be able to replicate what China is doing today and develop comprehensively in high-end technology,

producing "Make in India 2055." In population size and industrial base, India by then will indeed have the strength to do so. China, however, is doing this today largely because a ban on sales forced its hand: without independent development there was no way forward. India is different. By the 2040s there will be two semiconductor industrial systems in the world, and more than two countries able to produce lithography machines, photoresists and the like, and those countries will find it very hard to combine to blockade India. Moreover, India's bourgeoisie, dominated by private financial cliques, may not be willing to take that road.

Conclusion

From the discussion above we can draw some preliminary judgements:

1. India and the EU lack natural resources, while other countries such as the US and Indonesia have insufficient populations. China is the country most likely in the world to establish a self-sufficient economy.
2. Owing to technological constraints, Africa in the future, even if it unites regionally — forming, say, an East African Community or West African Community like the EU — will find it almost impossible to take the road of self-sufficiency.
3. China's population decline means that in 2040, the very year it completes its catch-up in semiconductors and large aircraft, it must already confront the problems brought by a large-scale population decline. By 2050 or 2060, China's employed population will very likely be insufficient to sustain a self-sufficient modern economy. Moreover, on its present development path, China has no thought at all of abandoning globalisation; on the contrary, it is accelerating its going-global, letting enterprises deploy worldwide and completing its own industrial upgrading in step.
4. India is the biggest variable, but the model dominated by India's private big capital will very likely make it even more globalised than China. For India's large enterprises to give up a global market of 9 billion people and operate only within a domestic market of one-odd billion is very hard to imagine.

The trend of de-globalisation, then, even if it develops, has limits; and in China and the other Southern countries the great trend is towards further globalisation, not de-globalisation. Indeed, as the negative effects of de-globalisation in the centre countries gradually appear, de-globalisation itself may well be reversed.

Notes

[1] Foreign direct investment, net inflows (BoP, current US\$)

中文原文: Foreign direct investment, net inflows (BoP, current US\$)

<https://data.worldbank.org/indicator/BX.KLT.DINV.CD.WD>

[2] Trade (% of GDP)

中文原文: Trade (% of GDP)

<https://data.worldbank.org/indicator/NE.TRD.GNFS.ZS>

[3] (source not preserved)

<https://zh.wikipedia.org/wiki/1939%E5%B9%B4%E5%BE%B7%E5%9C%8B%E4%BA%BA%E5%8F%A3%E6%99%AE%E6%9F%A5>

[4] US Population From 1900

中文原文: US Population From 1900

<http://demographia.com/db-uspop1900.htm>

[5] (source not preserved)

<https://www.nber.org/system/files/chapters/c3136/c3136.pdf>

[6] Demographic implications of autocentric development

中文原文: Demographic implications of autocentric development

<https://scispace.com/pdf/demographic-implications-of-autocentric-development-5dj5ne4ln2.pdf>

[7] (source not preserved)

<https://zh.wikipedia.org/wiki/%E6%97%A5%E6%9C%AC%E4%BA%BA%E5%8F%A3>

[8] Soviet Union (276 million), Hungary (10 million), Bulgaria (8.9 million), Czechoslovakia (15 million), Poland (37 million), East Germany (16 million), Romania (22 million)

中文原文: 苏联 (2.76亿), 匈牙利 (1000万), 保加利亚 (890万), 捷克斯洛伐克 (1500万), 波兰 (3700万), 东德 (1600万), 罗马尼亚 (2200万)

[9] (source not preserved)

<https://geofactbook.com/fact/total-population/1985>

[10] "Chinese-Style Modernization, Common Prosperity and the Future Development of Agriculture" — coupling and matching with Japan's development path, the number employed in the primary sector will fall from 177.15 million in 2020 to 78.5932 million in 2035, of which the number of "full-time workers" needed by agriculture will fall from 111.75 million in 2020 to 49.5801 million in 2035; by 2050, on a pessimistic assumption, primary-sector employment will fall to 38.5216 million, of which the "full-time workers" needed by agriculture will fall to 24.3012 million; even on an optimistic assumption, primary-sector employment will fall to 40.6598 million, of which agricultural employment will fall to 25.6499 million (see Table 5)

中文原文: 《中国式现代化、共同富裕与未来农业发展》耦合匹配日本的发展路径, 2035年第一产业就业人数将从2020年的17715万下降为2035年的7859.32万, 其中农业所需“全职劳动者”人数将从2020年的11175万下降为2035年的4958.01万; 到2050年, 如按悲观假设预测, 第一产业就业人数将下降为3852.16万, 其中农业所需“全职劳动者”人数将下降为2430.12万; 即使按乐观假设预测, 第一产业就业人数也会下降为4065.98万, 其中农业的就业人数也将下降为2564.99万 (详见表5)

<http://hnxb.cnjournals.net/hznydxsk/article/html/202501005?st=search>

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